Atlas Orrery - Technical Architecture

> Technical architecture, ownership boundaries, runtime structure, and contracts.

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PDF Export Profile (A4 Portrait)

Setting	Value
Page size	A4 Portrait
Margin	20 mm
Header	Left: `Atlas Orrery` | Right: `Technical Architecture`
Footer	`Atlas Orrery – Technical Architecture | <page>`
Title size	24-28 pt, bold
Section heading	15-17 pt, bold
Subheading	12-13 pt, semibold
Body	10.5-11.5 pt
Code block	9.5-10 pt, monospaced
Line spacing	1.25-1.35
Body font	Inter / Aptos / Calibri / IBM Plex Sans
Code font	JetBrains Mono / Consolas / Courier New

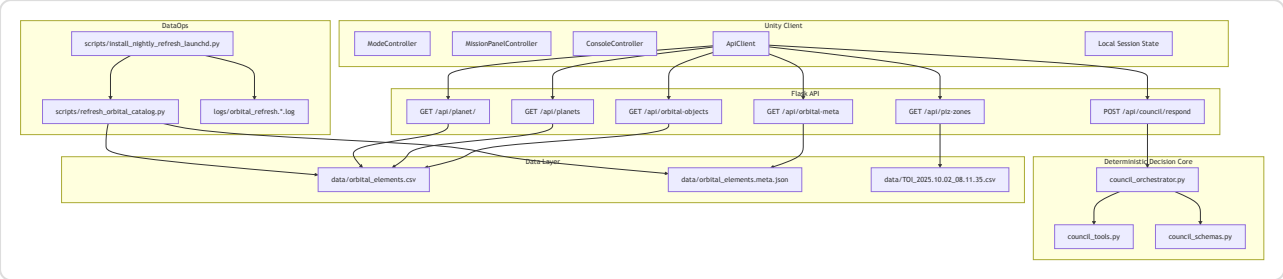
Diagram Export Rule

- Render Mermaid diagrams to SVG/PNG before final PDF export.
- Keep one diagram per concern (system topology vs runtime request flow).
- Use full-width diagrams, no raw Mermaid in final submission PDF.

What This Document Establishes

- Runtime decision ownership: deterministic council core in backend.
- Contract ownership: canonical normalization and response schema in `council_schemas.py`.
- Dataset publication ownership: refresh scripts publish runtime artifacts.
- Module boundaries and dependency direction across Unity, Flask, core, data, and ops.
- Verification mapping from architecture concerns to test and operational gates.

1) System Architecture Overview



Atlas Orrery runs as a single runtime stack: Unity controls interaction, Flask exposes stable HTTP boundaries, and deterministic backend logic produces council decisions from a refreshed orbital catalog.

2) Runtime Architecture

Client

- `ModeController` manages mode state (`sandbox`, `challenge`, `discovery`).
- `MissionPanelController` renders `headline`, recommendation, and player options.
- `ConsoleController` renders council vote timeline and caution context.
- `ApiClient` owns council/catalog API calls and request lifecycle handling.
- Local session state carries the active context between interactions.

Backend API

- Flask route layer is the only transport boundary between Unity and decision core.
- `POST /api/council/respond` is the primary decision entrypoint.
- Catalog and detail endpoints provide supporting read paths for UI and context exploration.

Decision Core

- `council_orchestrator.py` composes status branch + recommendation package.
- `council_tools.py` owns deterministic ranking, scoring, and vote building.
- `council_schemas.py` normalizes incoming payload and enforces contract shape.

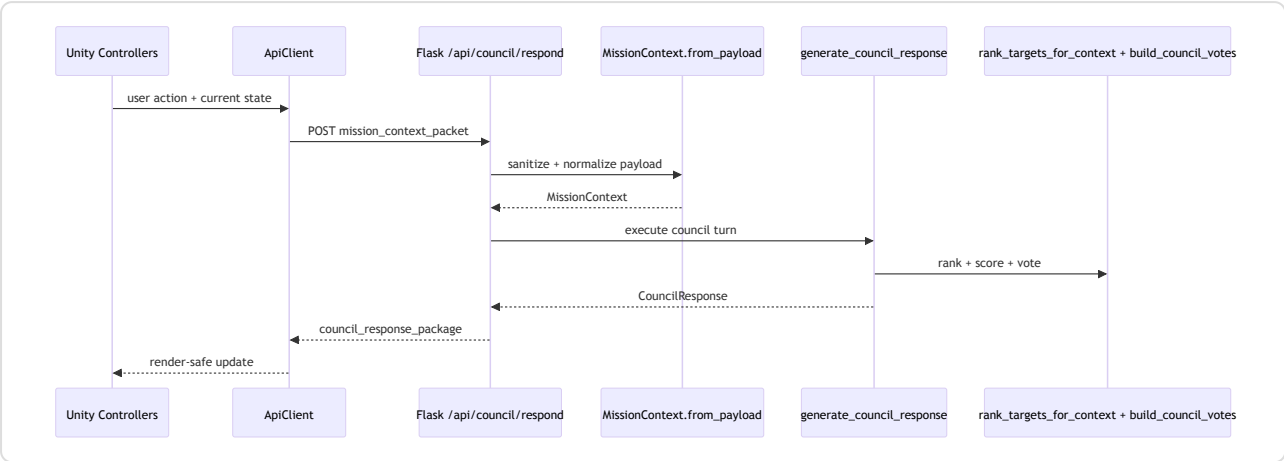
Data Layer

- `data/orbital_elements.csv`: runtime orbital catalog artifact.
- `data/orbital_elements.meta.json`: source/refresh metadata for runtime checks.
- `data/TOI_2025.10.02_08.11.35.csv`: supporting exploration dataset for PIZ endpoint.

Auxiliary Endpoints / Supporting Data Services

- `GET /api/piz-zones` is an auxiliary exploration endpoint backed by TOI data.
- It is not part of the council decision loop and does not affect council branch status.

3) Runtime Request Flow



- Request boundary: `POST /api/council/respond`.
- Parsing boundary: JSON parse + `MissionContext.from_payload` normalization.
- Decision boundary: `generate_council_response` orchestration.
- Deterministic computation boundary: `rank_targets_for_context`, `build_council_votes`.
- Response boundary: stable contract keys for safe UI rendering.

4) Code Map and Dependency Direction

Backend

- `server.py`
- Owns Flask routes, endpoint transport, data loading, and catalog object building.
- Depends on `generate_council_response` for mission response path.
- `council_orchestrator.py`
- Depends on schema types from `council_schemas.py`.
- Depends on deterministic primitives from `council_tools.py`.
- `council_tools.py`
- Deterministic compute layer only.
- No Flask dependency.
- No network dependency.
- `council_schemas.py`
- Canonical schema/normalization layer.
- Shared by request parsing and response assembly.

Scripts

- `scripts/refresh_orbital_catalog.py`: refresh and publish orbital artifacts.
- `scripts/install_nightly_refresh_launchd.py`: local scheduling and job wiring.

Tests

- `test_council_orchestrator.py`: validates branch behavior and response shape.

Dependency Rules

- Unity does not own scientific scoring.

- Flask transport layer does not own decision logic.
- Decision core does not depend on network calls at runtime.
- Schema layer is the contract source for client-backend stability.

Implementation Anchors

- `server.py` owns HTTP boundary and dispatch.
- `council_orchestrator.py` owns branch synthesis (`candidate_found`, `candidate_with_risk`, `insufficient_evidence`).
- `council_tools.py` owns deterministic score/rank/vote functions.
- `council_schemas.py` owns normalization and contract dataclasses.
- `test_council_orchestrator.py` verifies branch behavior in executable tests.

5) API Surface

Council

- `POST /api/council/respond`

Catalog

- `GET /api/orbital-objects`
- `GET /api/orbital-meta`
- `GET /api/planets` (legacy projection)
- `GET /api/planet/<planet_id>`

Auxiliary

- `GET /api/piz-zones`

6) Technical Contracts

Input Contract (`mission_context_packet`)

```
{
  "mode": "challenge",
  "player_goal": "find high-potential habitable candidates in 5 minutes",
  "selected_planet_id": "Kepler-442 b",
  "selected_piz_id": null,
  "filters": {
    "showConfirmed": true,
    "showHabitable": true,
    "radiusMin": 0.7,
    "radiusMax": 2.2,
    "periodMin": 1,
    "periodMax": 500
  },
  "challenge_state": {
    "active": true,
    "objective": "Find 2 candidate worlds",
    "progress": 1
  },
  "recent_actions": ["spiral_scan", "open_planet_modal"]
}
```

Output Contract (`council_response_package`)

```
{
  "mission_status": "candidate_with_risk",
```

```
  "headline": "Council uu tien Kepler-442 b cho buoc ke tiep",
  "primary_recommendation": {
    "action": "targeted_scan",
    "target_id": "Kepler-442 b",
    "reason": "Scored 0.81 on baseline habitability under current goal"
  },
  "council_votes": [
    {
      "agent": "Navigator",
      "stance": "support",
      "confidence": 0.82,
      "message": "Recommend targeted follow-up based on ranking gain.",
      "evidence_fields": ["pl_orbper", "pl_orbsmax", "sy_dist"]
    },
    {
      "agent": "Astrobiologist",
      "stance": "support",
      "confidence": 0.8,
      "message": "Radius-temperature-insolation triad is within exploratory viability bounds.",
      "evidence_fields": ["pl_rade", "pl_eqt", "pl_insol"]
    },
    {
      "agent": "Climate",
      "stance": "caution",
      "confidence": 0.71,
      "message": "Orbital uncertainty needs deeper verification.",
      "evidence_fields": ["pl_orbeccen", "pl_orbper", "pl_orbincl"]
    }
  ],
  "player_options": [
    "Run targeted scan",
    "Compare nearest analogs",
    "Open full data dossier"
  ],
  "discovery_log_entry": "Kepler-442 b promoted after council triage.",
  "evidence_summary": {
    "radius_earth": 1.34,
    "temp_k": 285.0,
    "insolation": 0.95,
    "eccentricity": 0.08,
    "period_days": 112.4
  }
}
```


Guardrails / Invariants

- `mode` accepted values: `sandbox`, `challenge`, `discovery`.
- Filter ranges are normalized; reversed min/max are corrected.
- `recent_actions` is normalized to capped `list[str]` (max 20 entries).
- Response keeps a stable keyset across all statuses for render safety.

Source of Truth Boundaries

- Dataset source of truth: published `orbital_elements.csv` artifact after refresh validation.
- Contract source of truth: `council_schemas.py`.
- Runtime decision source of truth: deterministic orchestrator + tools layer.

7) Responsibility Boundaries

Component	Owns	Out of scope
Unity controllers + ApiClient	Interaction state, API calls, rendering	Scientific scoring and dataset mutation
Flask route layer	HTTP boundary and transport response	Decision scoring logic
`council_orchestrator.py`	Branch synthesis and response composition	Dataset file lifecycle

Component	Owns	Out of scope
`council_tools.py`	Ranking/scoring/voting compute	Network transport and persistence
`council_schemas.py`	Payload normalization and contract typing	Endpoint routing
Refresh scripts	Dataset publication and metadata update	Runtime council decision path

8) Deployment and Runtime Assumptions

Assumption Boundaries

Boundary	Statement
Guaranteed	Runtime council path is deterministic and does not require external model/network calls; Unity receives stable response keys across statuses.
Assumed	Runtime catalog size remains within demo target bounds; refresh completes before demo window; backend runs as a local single-instance process.
Out of scope	Horizontal scaling, distributed orchestration, multi-region failover.

Non-goals

- No distributed architecture in MVP.
- No real-time stream processing for catalog ingestion.
- No scientific scoring logic in Unity.
- No LLM replacement of deterministic ranking core in MVP.

Key Architectural Decisions

Decision	Why	Trade-off
Deterministic council core	Stable, explainable, testable demo behavior	Less expressive than full LLM-first reasoning
Single-instance local Flask runtime	Lower setup cost and smaller failure domain	Not intended for production-scale traffic
Contract ownership in `council_schemas.py`	Reduces client-backend drift	Requires strict schema governance during iteration
Artifact snapshot before demo	Minimizes late refresh risk	Data may not be latest available at demo time
Separate architecture and pipeline docs	Clean separation of design vs execution control	Reviewers must read both docs for full system view

9) NFR and SLO for Hackathon Demo

Performance

- `POST /api/council/respond` p95 < 1200 ms (local).
- Ranking p95 < 120 ms for runtime catalog <= 900 objects.

Reliability

- Invalid payload is normalized instead of crashing endpoint.
- Empty candidate set degrades to `insufficient_evidence` response.

Security

- No hardcoded secrets in repository.
- Credentials (future model extensions) handled via `.env`.

Observability

- Runtime logs include at least `request_id`, `mode`, `candidate_count`, `mission_status`, `latency_ms`.
- Refresh logs are isolated in `logs/orbital_refresh.*.log`.

10) Architectural Risks and Mitigations

1. Contract drift between Unity and backend. - Mitigation: schema-centric contracts in `council_schemas.py` + contract checks.
2. Refresh failure close to demo time. - Mitigation: pre-demo refresh window + artifact freeze.
3. Dead-end user flows due to narrow filters. - Mitigation: explicit `insufficient_evidence` branch with actionable options.
4. Cold-start latency spike on first mission request. - Mitigation: warm catalog cache before live demo flow.

Verification Mapping

Concern	Verified by
Branch correctness	`test_council_orchestrator.py`
Contract stability	schema guardrails + contract checks in pipeline gates
Dataset validity	refresh validation before artifact publish
Runtime readiness	API smoke checks + rehearsal pass
Performance target	latency verification against demo SLO

11) Evolution Path

MVP

- Deterministic council loop with stable contracts.

Hybrid

- Model-assisted explanation layer after deterministic decision output.
- Deterministic ranking core remains canonical.

Adaptive

- Recommendation tuning based on user behavior and acceptance patterns.

12) Conclusion

This architecture is intentionally demo-first: clear ownership boundaries, deterministic decisioning, stable contracts for Unity rendering, and controlled data publication to keep runtime behavior predictable under hackathon constraints.